

Inner Products, Eigenvectors, and the Spectral Theorem

Quantum mechanics, four parts early, with the physics stripped off.

WHERE WE ARE

Here is the stake, stated as plainly as it can be stated: **every postulate of quantum mechanics is a statement about Hermitian operators on an inner-product space.** Not *modelled by*, not *analogous to* — is. Measurement outcomes are eigenvalues. States are unit vectors. Time evolution is a unitary map. Compatible observables are commuting operators. Take the physics away and what is left is a chapter of linear algebra, and it is this one.

So we are going to prove that chapter of linear algebra now, four parts early, with no physics attached whatsoever. No wavefunctions, no $\hbar$, no measurement. By the end you will have *proved* — with no mention of physics anywhere in the argument — that a certain class of operators has only real eigenvalues, and that eigenvectors belonging to different eigenvalues are exactly perpendicular. When Chapter 4.2 announces that measurement results are real numbers and that distinct outcomes are perfectly distinguishable, you will already have the proof. Chapter 4.2 becomes a *translation exercise*: a table of renamings, not a new subject.

That is the whole design of this book compressed into one chapter. The mathematics is not scaffolding erected to support the physics. It is the physics, wearing different words.

TOOLS YOU'LL NEED — Chapter 0.1: nothing beyond fluency. Chapter 0.3: Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$, and power series. Chapter 0.4: vector spaces, basis and dimension, linear maps and their matrices, change of basis, determinant, trace, and the fact that over $\mathbb{C}$ every square matrix has at least one eigenvalue. Everything here is built on those.

1 · Inner products

A vector space as Chapter 0.4 left it is a bleak place. You can add vectors and scale them, and that is all. There is no length, no angle, no notion of two vectors pointing in different directions. Every geometric word you know is missing. An **inner product** is the single extra structure that puts all of them back at once.

Why the axioms look the way they do

In $\mathbb{R}^n$ you already have one: the dot product $\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$, whose essential property is that $\mathbf{v} \cdot \mathbf{v} = \sum_i v_i^2$ is the squared length — a non-negative number that vanishes only for the zero vector. Try to carry that over to $\mathbb{C}^n$ unchanged and it breaks immediately. Take $v = (1, i)$ in $\mathbb{C}^2$:

$$\sum_i v_i^2 = 1^2 + i^2 = 1 - 1 = 0, \quad (0.5.1)$$

and yet $v \neq 0$. A non-zero vector with zero length is useless; we would lose the ability to say that two states are different. The repair is forced, and it is the only repair available: replace v_i^2 by $|v_i|^2 = \bar{v}_i v_i$, which is non-negative for every complex number and zero only when $v_i = 0$. So the complex inner product must conjugate one of its two arguments. Everything below is bookkeeping around that single fact.

An **inner product** on a complex vector space V is a map $V \times V \rightarrow \mathbb{C}$, written $\langle u, v \rangle$, satisfying three axioms.

$$\begin{aligned} \text{(i) conjugate symmetry:} \quad & \langle u, v \rangle = \overline{\langle v, u \rangle}, \\ \text{(ii) linearity in slot 2:} \quad & \langle u, \alpha v + \beta w \rangle = \alpha \langle u, v \rangle + \beta \langle u, w \rangle, \\ \text{(iii) positive definiteness:} \quad & \langle v, v \rangle > 0 \text{ for all } v \neq 0. \end{aligned} \quad (0.5.2)$$

Three remarks, in increasing order of importance.

Axiom (iii) is not quite well posed until (i) is used. " $\langle v, v \rangle > 0$ " presumes $\langle v, v \rangle$ is real, and that is a *consequence*, not an assumption: setting $u = v$ in (i) gives $\langle v, v \rangle = \overline{\langle v, v \rangle}$, and a complex number equal to its own conjugate is real. Good — the axioms are consistent, and we derived the fact rather than assuming it.

Linearity in one slot forces conjugate-linearity in the other. We only demanded linearity in the second argument. The first is then determined:

$$\langle \alpha u, v \rangle = \overline{\langle v, \alpha u \rangle} = \overline{\alpha \langle v, u \rangle} = \bar{\alpha} \overline{\langle v, u \rangle} = \bar{\alpha} \langle u, v \rangle. \quad (0.5.3)$$

Pull a scalar out of the *second* slot and it comes out unchanged; pull it out of the *first* and it comes out conjugated. This asymmetry is real and you must keep track of it. It is also the reason the notation of §2 exists.

The convention. Which slot is the linear one is pure convention, and the two communities chose differently.

Mathematicians make the first slot linear; physicists make the second. We have chosen the physics convention above, and will keep it for the whole book, because in Chapter 4.2 the object $\langle \phi, \psi \rangle$ becomes a probability amplitude, read right-to-left as "amplitude to find ψ in state ϕ ", and the linear slot had better be the one holding the state that is actually evolving. If you open a mathematics text and every formula appears to have its conjugates on the wrong side, this is why.

The standard examples

SPACE	INNER PRODUCT	WHERE IT TURNS UP
$\mathbb{R}^n$	$\sum_i u_i v_i$	Ordinary geometry; least squares (§3)
$\mathbb{C}^n$	$\sum_i \bar{u}_i v_i$	Spin, qubits, any finite quantum system (4.7)
$\mathbb{R}^n$, weighted	$\sum_i w_i u_i v_i, w_i > 0$	Weighted regression; χ^2 fitting
Functions on $[a, b]$	$\int_a^b \overline{f(x)} g(x) dx$	Fourier series (0.9); wavefunctions (4.3)
$n \times n$ matrices	$\text{tr}(A^\dagger B)$	Density matrices (4.8)

Check the fourth row against the axioms and you will find nothing new is needed: conjugate symmetry because $\overline{\int \bar{g} f} = \int \bar{f} g$, linearity in the second slot because integration is linear, positive definiteness because $\int |f|^2 \geq 0$ with equality only if $f \equiv 0$. That row is the entire reason Chapter 4.3 is possible: *a space of functions is an inner-product space*, and every theorem of this chapter will try to move there. Some of them survive the trip and some need repair, which is the subject of a warning later on.

Length, distance, and the angle we cannot yet define

Given an inner product, define the **norm** and the **distance**:

$$\|v\| \equiv \sqrt{\langle v, v \rangle}, \quad d(u, v) \equiv \|u - v\|. \quad (0.5.4)$$

Both are well defined: $\langle v, v \rangle$ is real and non-negative, so the square root is real. Call u and v **orthogonal** when $\langle u, v \rangle = 0$, and note that this is symmetric, since $\langle v, u \rangle = \overline{\langle u, v \rangle} = 0$ too. Expanding a norm-squared using the axioms gives the generalised Pythagoras:

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle, \quad (0.5.5)$$

because $\langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$ and the middle two are conjugates of each other, so they add to twice the real part. When $u \perp v$ the cross term dies and $\|u + v\|^2 = \|u\|^2 + \|v\|^2$: Pythagoras, in any dimension, over $\mathbb{C}$, derived in one line from three axioms.

Now the angle. In $\mathbb{R}^n$ you were told $\cos \theta = \mathbf{u} \cdot \mathbf{v} / (\|u\| \|v\|)$. We would like to *define* the angle that way in general — but a definition is only legal if the right-hand side lands in $[-1, 1]$, and nothing so far guarantees it. Proving that it does is the next item, and it is much more than a technicality.

The Cauchy–Schwarz inequality

The claim is that for all u, v in an inner-product space,

$$\boxed{|\langle u, v \rangle|^2 \leq \langle u, u \rangle \langle v, v \rangle} \quad (0.5.6)$$

with equality exactly when u and v are parallel. The proof is one idea: *positivity of a norm is an infinite supply of inequalities, and we get to choose which one to use.* Whatever $\lambda \in \mathbb{C}$ we pick, axiom (iii) guarantees

$$0 \leq \langle u - \lambda v, u - \lambda v \rangle. \quad (0.5.7)$$

Expand it, remembering that the first slot conjugates by (0.5.3) and the second does not:

$$\begin{aligned} \langle u - \lambda v, u - \lambda v \rangle &= \langle u, u \rangle - \lambda \langle u, v \rangle - \bar{\lambda} \langle v, u \rangle + |\lambda|^2 \langle v, v \rangle \\ &= \langle u, u \rangle - 2 \operatorname{Re}(\lambda \langle u, v \rangle) + |\lambda|^2 \langle v, v \rangle, \end{aligned} \quad (0.5.8)$$

where the two middle terms combined because $\bar{\lambda} \langle v, u \rangle = \overline{\lambda \langle u, v \rangle}$. This is non-negative for *every* λ , so it is non-negative for the λ that makes it smallest, and we now choose that one. Assume $v \neq 0$ (if $v = 0$ both sides of (0.5.6) are zero and there is nothing to prove) and take

$$\lambda = \frac{\langle v, u \rangle}{\langle v, v \rangle} = \frac{\overline{\langle u, v \rangle}}{\langle v, v \rangle}. \quad (0.5.9)$$

With that choice $\lambda \langle u, v \rangle = |\langle u, v \rangle|^2 / \langle v, v \rangle$, which is real and non-negative, and $|\lambda|^2 \langle v, v \rangle = |\langle u, v \rangle|^2 / \langle v, v \rangle$ as well. Substituting into (0.5.8), the two copies partially cancel:

$$0 \leq \langle u, u \rangle - \frac{2 |\langle u, v \rangle|^2}{\langle v, v \rangle} + \frac{|\langle u, v \rangle|^2}{\langle v, v \rangle} = \langle u, u \rangle - \frac{|\langle u, v \rangle|^2}{\langle v, v \rangle}. \quad (0.5.10)$$

Multiply through by $\langle v, v \rangle > 0$ and you have (0.5.6). Equality holds precisely when the quantity in (0.5.7) is zero, i.e. when $u = \lambda v$ — the two vectors are parallel.

So the angle is now legal. Dividing (0.5.6) by $\|u\|^2 \|v\|^2$ shows $|\langle u, v \rangle| / (\|u\| \|v\|) \leq 1$, and we may define

$$\cos \theta \equiv \frac{\operatorname{Re} \langle u, v \rangle}{\|u\| \|v\|} \in [-1, 1]. \quad (0.5.11)$$

▼ Grind box — the triangle inequality, and what λ was

The triangle inequality falls out. Start from (0.5.5) and bound the cross term using $\operatorname{Re} z \leq |z|$ and then Cauchy–Schwarz:

$$\begin{aligned}\|u + v\|^2 &= \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle \\ &\leq \|u\|^2 + \|v\|^2 + 2 \|u\| \|v\| = (\|u\| + \|v\|)^2.\end{aligned}$$

Taking square roots, $\|u + v\| \leq \|u\| + \|v\|$. Every metric-space fact you have ever used about these spaces descends from this line, and this line descends from (0.5.6).

What the magic λ was. It looks pulled from a hat. It is not: minimise (0.5.8) honestly. Write $\lambda = t \overline{\langle u, v \rangle}$ with t real, so that $\lambda \langle u, v \rangle = t |\langle u, v \rangle|^2$ is real. Then the expression is the real quadratic

$$q(t) = \langle u, u \rangle - 2t |\langle u, v \rangle|^2 + t^2 |\langle u, v \rangle|^2 \langle v, v \rangle,$$

$$q'(t) = 0 \implies t = \frac{1}{\langle v, v \rangle},$$

which reproduces (0.5.9) exactly. And the minimising vector $\lambda v = \frac{\langle v, u \rangle}{\langle v, v \rangle} v$ is precisely the *orthogonal projection of u onto the line through v* , which is §3. Cauchy–Schwarz is the statement that a projection is never longer than the thing being projected. Everything in this chapter is one geometric picture seen from different angles.

An alternative proof, for the file. If $\|u\| = \|v\| = 1$ then $0 \leq \|u - e^{i\varphi} v\|^2 = 2 - 2 \operatorname{Re}(e^{-i\varphi} \langle u, v \rangle)$; choosing φ to be the argument of $\langle u, v \rangle$ makes the real part equal $|\langle u, v \rangle|$ and gives $|\langle u, v \rangle| \leq 1$ directly. The general case follows by rescaling. Same idea, phase instead of magnitude.

THIS INEQUALITY IS THE UNCERTAINTY PRINCIPLE

Chapter 4.6 will prove that for any two observables A and B measured on a state ψ ,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|.$$

The proof there is (0.5.6) applied to the two vectors

$$u = (A - \langle A \rangle)\psi, \quad v = (B - \langle B \rangle)\psi,$$

followed by one line splitting $\langle u, v \rangle$ into its real and imaginary parts. That is the entire derivation.

Nothing is added in Chapter 4.6 except the physical meaning of the symbols. The Heisenberg uncertainty principle — the single most quoted statement in twentieth-century physics, the one that gets prose written about the limits of human knowledge — is Cauchy–Schwarz. You have just proved it. You are simply not yet allowed to say what the letters mean.

2 · Orthonormal bases

Chapter 0.4 showed that any basis lets you write every vector uniquely in coordinates. It also showed that finding those coordinates means solving a linear system, which is work. With an inner product, one class of basis makes that work vanish.

A set $\{e_1, \dots, e_n\}$ is **orthonormal** if

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j. \end{cases} \quad (0.5.12)$$

Orthonormal sets are automatically linearly independent, so in dimension n any n of them form a basis. (Suppose $\sum_i c_i e_i = 0$. Take $\langle e_j, \cdot \rangle$ of both sides: linearity in the second slot gives $\sum_i c_i \langle e_j, e_i \rangle = c_j = 0$ for every j . Independence, in one line.)

That same computation is the payoff. Write any $v = \sum_i c_i e_i$ and hit it with $\langle e_j, \cdot \rangle$:

$$\langle e_j, v \rangle = \left\langle e_j, \sum_i c_i e_i \right\rangle = \sum_i c_i \delta_{ji} = c_j, \quad (0.5.13)$$

and therefore

$$\boxed{v = \sum_i \langle e_i, v \rangle e_i} \quad (0.5.14)$$

Coordinates have become inner products. No system to solve, no matrix to invert — to find the i -th coordinate you take one inner product. This is the single practical reason orthonormal bases dominate physics, and notice that the physics convention is paying already: because the second slot is the linear one, $\langle e_i, v \rangle$ is the coefficient with no stray conjugate.

Two more consequences, each obtained by substituting (0.5.14) into an inner product and using (0.5.12) to collapse the double sum:

$$\langle u, v \rangle = \sum_i \overline{\langle e_i, u \rangle} \langle e_i, v \rangle, \quad \|v\|^2 = \sum_i |\langle e_i, v \rangle|^2. \quad (0.5.15)$$

The second is **Parseval's identity**: the squared length of a vector is the sum of the squared magnitudes of its coordinates — but only in an orthonormal basis. In Chapter 0.9 the e_i become the Fourier modes and this becomes the statement that total energy equals the sum of energies in each mode; in Chapter 4.2 the $|\langle e_i, v \rangle|^2$ become probabilities and this becomes the statement that they add to one.

Gram–Schmidt: orthonormal bases always exist

The formulae above are worthless if orthonormal bases are rare. They are not. Given *any* basis $\{v_1, \dots, v_n\}$, the following procedure manufactures an orthonormal one. Set

$$w_k = v_k - \sum_{i < k} \langle e_i, v_k \rangle e_i, \quad e_k = \frac{w_k}{\|w_k\|}, \quad (0.5.16)$$

running $k = 1, 2, \dots, n$ in order (for $k = 1$ the sum is empty, so $e_1 = v_1 / \|v_1\|$). In words: *take the next vector, subtract off everything it has in common with the directions already fixed, and normalise what is left.*

▼ Grind box — why Gram–Schmidt works, line by line

Three things must be checked: that we never divide by zero, that the output is orthonormal, and that it spans the same space. All three come from one induction hypothesis:

$$H_k: \{e_1, \dots, e_k\} \text{ is orthonormal and } \text{span}\{e_1, \dots, e_k\} = \text{span}\{v_1, \dots, v_k\}.$$

Base case. $v_1 \neq 0$ because it belongs to a basis, so $\|v_1\| \neq 0$ and $e_1 = v_1 / \|v_1\|$ is a unit vector spanning the same line. H_1 holds.

No division by zero. Assume H_{k-1} . If w_k were 0, then by (0.5.16) $v_k = \sum_{i < k} \langle e_i, v_k \rangle e_i$ would lie in $\text{span}\{e_1, \dots, e_{k-1}\} = \text{span}\{v_1, \dots, v_{k-1}\}$, contradicting the linear independence of the v 's. So $w_k \neq 0$ and e_k is defined.

Orthogonality. For any $j < k$, take the inner product of (0.5.16) with e_j and use $\langle e_j, e_i \rangle = \delta_{ji}$:

$$\langle e_j, w_k \rangle = \langle e_j, v_k \rangle - \sum_{i < k} \langle e_i, v_k \rangle \delta_{ji} = \langle e_j, v_k \rangle - \langle e_j, v_k \rangle = 0.$$

Exactly one term of the sum survives, and it is exactly the term that cancels. Dividing by $\|w_k\|$ preserves this, so $e_k \perp e_j$ for all $j < k$, and $\|e_k\| = 1$ by construction.

Same span. From (0.5.16), e_k is a combination of v_k and earlier e 's, hence of $v_1, \dots, v_k$; and rearranged, v_k is a combination of $e_1, \dots, e_k$. So each spans the other's span, and H_k holds. Induction completes the proof at $k = n$. ■

Two consequences worth naming. First, *every* finite-dimensional inner-product space has an orthonormal basis — the theorems below never have to assume one exists. Second, Gram–Schmidt is *triangular*: e_k involves only $v_1, \dots, v_k$. Writing that relationship as a matrix statement gives $V = QR$ with Q having orthonormal columns and R upper triangular — the QR decomposition, which is how a computer actually solves the least-squares problem of §3.

Dirac notation, introduced where it is harmless

We now install the notation that Chapter 4.2 will use for everything. It is standard to meet it in the middle of learning quantum mechanics, where it acquires an entirely undeserved mystique. It is a piece of pure linear algebra and there is nothing in it. Here it is.

Write a vector $v \in V$ as $|v\rangle$, a **ket**. Now fix a vector u and consider the map that sends any w to the number $\langle u, w \rangle$. By axiom (ii) that map is *linear* in w — it is a **linear functional**, an element of the space V^* of linear maps $V \rightarrow \mathbb{C}$ (itself a vector space, since sums and scalar multiples of linear maps are linear). Call it $\langle u|$, a **bra**. Then:

$$\langle u| \text{ applied to } |v\rangle \equiv \langle u|v\rangle = \langle u, v \rangle \in \mathbb{C}. \quad (0.5.17)$$

The bracket closes, and a closed bracket is a *number*. Now leave it open. Define $|u\rangle \langle v|$ to be the map that acts on $|w\rangle$ by applying $\langle v|$ first and using the resulting number to scale $|u\rangle$:

$$(|u\rangle \langle v|) |w\rangle \equiv |u\rangle \langle v|w\rangle = \langle v, w \rangle u. \quad (0.5.18)$$

Is that a linear map? Check: $w \mapsto \langle v, w \rangle u$ is linear in w because $\langle v, \cdot \rangle$ is, so yes. The **outer product** $|u\rangle \langle v|$ **is an operator**; the **inner product** $\langle u|v\rangle$ **is a number**. In coordinates with respect to an orthonormal basis the distinction is the one you already know from Chapter 0.4:

$$\underbrace{\begin{pmatrix} \bar{u}_1 & \cdots & \bar{u}_n \end{pmatrix}}_{\langle u| \text{ (row)}} \underbrace{\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}}_{|v\rangle \text{ (column)}} = \text{a } 1 \times 1, \quad |u\rangle \langle v| = \text{an } n \times n. \quad (0.5.19)$$

Row times column is a number; column times row is a matrix. That is the whole content. Dirac notation is a device for making the *type* of an object visible in its *shape*: you can tell at a glance whether an expression is a scalar or an operator, without tracking any dimensions. Note also, from (0.5.3), that the map $u \mapsto \langle u|$ is conjugate-linear: $\langle \alpha u| = \bar{\alpha} \langle u|$. The conjugates in (0.5.19) are that fact in coordinates.

Completeness: the resolution of the identity

Now rewrite the expansion (0.5.14) in the new notation. Its coefficient $\langle e_i, v \rangle$ is $\langle e_i|v\rangle$, so

$$|v\rangle = \sum_i |e_i\rangle \langle e_i|v\rangle = \left(\sum_i |e_i\rangle \langle e_i| \right) |v\rangle. \quad (0.5.20)$$

The regrouping in the second step is legal by (0.5.18). Since this holds for *every* $|v\rangle$, the operator in brackets acts as the identity on every vector, and two linear maps that agree on every vector are equal:

$$\boxed{\sum_i |e_i\rangle \langle e_i| = \hat{I}} \quad (0.5.21)$$

This is the **completeness relation**, or *resolution of the identity*. It says the basis is not missing any directions: reconstruct any vector from its components and you get the vector back, with nothing left over. It is used constantly and always the same way — you insert $\hat{I}$ in the middle of an expression in the form (0.5.21), and a hard object turns into a sum of easy ones. Chapter 4.2 does this on nearly every page; so does Chapter 0.9, where the sum becomes an integral over a continuum of modes.

3 · Projection, and why least squares is a right angle

Let $W \subseteq V$ be a **subspace** — a subset closed under addition and scalar multiplication, hence a vector space in its own right, and one that inherits V 's inner product unchanged. By Gram–Schmidt it has an orthonormal basis $\{f_1, \dots, f_m\}$. Define the **orthogonal projection onto W** :

$$P = \sum_{j=1}^m |f_j\rangle \langle f_j|, \quad \text{so} \quad Pv = \sum_j \langle f_j, v \rangle f_j. \quad (0.5.22)$$

Compare with (0.5.21): this is the same construction, truncated. We keep only the directions inside W and throw the rest away. Three properties, all derived.

P lands in W and fixes W . Pv is visibly a combination of the f_j , so $Pv \in W$. And if $w \in W$ then (0.5.14) applied inside W says $w = \sum_j \langle f_j, w \rangle f_j = Pw$. Hence for any v , $Pv \in W$ gives $P(Pv) = Pv$:

$$P^2 = P. \quad (0.5.23)$$

Projecting something that has already been projected does nothing. That is what "projection" means, and (0.5.23) is the algebraic form of it.

P is symmetric under the inner product. Expanding both sides with (0.5.22),

$$\langle Pu, v \rangle = \sum_j \langle u, f_j \rangle \langle f_j, v \rangle = \langle u, Pv \rangle. \quad (0.5.24)$$

(The first equality uses conjugate-linearity in slot 1: $\overline{\langle f_j, u \rangle} = \langle u, f_j \rangle$.) In §4 this will be written $P^\dagger = P$.

The residual is orthogonal to W . For each j ,

$$\langle f_j, v - Pv \rangle = \langle f_j, v \rangle - \langle f_j, v \rangle = 0, \quad (0.5.25)$$

so $v - Pv \perp f_j$ for every j , hence $v - Pv \perp W$ entirely (an inner product with any combination of the f_j is the corresponding combination of zeros).

Projection is the closest point — proved

Now the theorem that makes projection useful rather than merely tidy.

Claim: for every $w \in W$, $\|v - w\| \geq \|v - Pv\|$, with equality only for $w = Pv$.

Split the vector $v - w$ into two pieces by adding and subtracting Pv :

$$v - w = \underbrace{(v - Pv)}_{\perp W} + \underbrace{(Pv - w)}_{\in W}. \quad (0.5.26)$$

The first piece is orthogonal to W by (0.5.25); the second lies in W because both Pv and w do. So the two pieces are orthogonal to each other, and Pythagoras (0.5.5) applies with no cross term:

$$\|v - w\|^2 = \|v - Pv\|^2 + \|Pv - w\|^2. \quad (0.5.27)$$

The first term does not depend on w ; the second is non-negative and vanishes only when $w = Pv$. So the minimum is attained there and nowhere else. ■

Read (0.5.27) once more. It says **"nearest point" and "perpendicular residual" are the same condition**. Not similar, not related: the same. Every optimisation problem that turns out to have a linear-algebra answer is this theorem in disguise.

YOU HAVE JUST DERIVED LEAST SQUARES

Fit $y \approx X\beta$, with $y \in \mathbb{R}^n$ the outcomes, X the $n \times p$ design matrix, and β the coefficients to be chosen. Minimising the residual sum of squares means minimising $\|y - X\beta\|^2$ — a squared distance in $\mathbb{R}^n$ with the ordinary inner product.

As β ranges over all of $\mathbb{R}^p$, the vector $X\beta$ ranges over exactly the *column space* of X : the subspace spanned by the predictors. So the problem is: find the point of that subspace closest to y . By the theorem just proved, the answer is the orthogonal projection, and it is characterised by the residual being perpendicular to the subspace — which means perpendicular to every column of X :

$$X^\top (y - X\hat{\beta}) = 0 \iff X^\top X\hat{\beta} = X^\top y.$$

Those are the **normal equations**, and the word "normal" in their name has always meant "perpendicular". Solving them gives $\hat{\beta} = (X^\top X)^{-1} X^\top y$ and $\hat{y} = Hy$ with $H = X(X^\top X)^{-1} X^\top$ — the "hat matrix" of every regression output you have ever read, which satisfies $H^2 = H$ and $H^\top = H$ because it is the projection P of (0.5.22).

One more line pays a dividend. If X contains an intercept column, the constant vector lies in the column space, so (0.5.27) applied to $v = y - \bar{y}\mathbf{1}$ gives

$$\underbrace{\|y - \bar{y}\mathbf{1}\|^2}_{\text{total SS}} = \underbrace{\|\hat{y} - \bar{y}\mathbf{1}\|^2}_{\text{explained SS}} + \underbrace{\|y - \hat{y}\|^2}_{\text{residual SS}}.$$

The ANOVA decomposition is Pythagoras' theorem, and R^2 is the squared cosine of the angle between the centred outcome vector and its projection. Regression is trigonometry in n dimensions.

4 • The adjoint

Every operator on an inner-product space has a partner, defined not by a formula in coordinates but by how it interacts with the inner product.

Definition. The **adjoint** $A^\dagger$ of a linear map $A : V \rightarrow V$ is the linear map satisfying

$$\langle Au, v \rangle = \langle u, A^\dagger v \rangle \quad \text{for all } u, v \in V. \quad (0.5.28)$$

Read it as a rule for *moving an operator across the comma*: it costs you a dagger. No basis appears anywhere in (0.5.28), which is the point — the adjoint is a property of A and the inner product, not of any coordinate system.

It is unique. Suppose B and C both satisfy (0.5.28). Then $\langle u, (B - C)v \rangle = 0$ for every u and v . Choose $u = (B - C)v$: this gives $\|(B - C)v\|^2 = 0$, so $(B - C)v = 0$ for every v , so $B = C$.

It exists. Fix an orthonormal basis and define $A^\dagger v \equiv \sum_i \langle Ae_i, v \rangle e_i$. This is linear in v (the inner product is linear in slot 2), and expanding u via (0.5.14),

$$\langle Au, v \rangle = \sum_i \overline{\langle e_i, u \rangle} \langle Ae_i, v \rangle = \left\langle u, \sum_i \langle Ae_i, v \rangle e_i \right\rangle, \quad (0.5.29)$$

which is $\langle u, A^\dagger v \rangle$. Existence and uniqueness settled, in five lines.

In an orthonormal basis it is the conjugate transpose

Chapter 0.4 defined the matrix of A by $Ae_j = \sum_i A_{ij} e_i$. Taking $\langle e_i, \cdot \rangle$ of that and using (0.5.12) gives the useful formula $A_{ij} = \langle e_i, Ae_j \rangle$: *a matrix element is an inner product*. Now compute the matrix of $A^\dagger$ the same way, using (0.5.28) and then conjugate symmetry:

$$(A^\dagger)_{ij} = \langle e_i, A^\dagger e_j \rangle = \langle Ae_i, e_j \rangle = \overline{\langle e_j, Ae_i \rangle} = \overline{A_{ji}}. \quad (0.5.30)$$

So $A^\dagger$ is the **conjugate transpose** — transpose the matrix and conjugate the entries. That familiar recipe is not the definition; it is what the definition (0.5.28) looks like once you commit to an orthonormal basis. Note the hypothesis: in a non-orthonormal basis (0.5.30) is simply false.

Three algebraic rules follow immediately from (0.5.28), each by moving operators across the comma one at a time:

$$(A^\dagger)^\dagger = A, \quad (\alpha A)^\dagger = \bar{\alpha} A^\dagger, \quad (AB)^\dagger = B^\dagger A^\dagger. \quad (0.5.31)$$

For the last: $\langle ABu, v \rangle = \langle Bu, A^\dagger v \rangle = \langle u, B^\dagger A^\dagger v \rangle$, and uniqueness does the rest. The order reverses, exactly as it does for the transpose and the inverse. One more, used constantly below and derived the same way: $(|u\rangle \langle v|)^\dagger = |v\rangle \langle u|$.

The three species of operator

NAME	CONDITION	WHAT IT IS FOR
Hermitian (self-adjoint)	$A^\dagger = A$	Observables. §6 is about these
Unitary	$U^\dagger U = \hat{I}$	Symmetries, time evolution, change of orthonormal basis
Normal	$AA^\dagger = A^\dagger A$	The largest class that can be diagonalised orthonormally

Both Hermitian and unitary operators are normal — for Hermitian because $AA^\dagger = A^2 = A^\dagger A$, for unitary because $U^\dagger U = \hat{I}$ implies (in finite dimensions) that U is invertible with $U^{-1} = U^\dagger$, whence $UU^\dagger = \hat{I}$ too. For a Hermitian matrix, (0.5.30) says $A_{ij} = \overline{A_{ji}}$: the diagonal entries are real, and entries across the diagonal are conjugates. A real Hermitian matrix is just a *symmetric* matrix.

Unitary maps preserve all geometry

This is a two-line calculation with large consequences. If $U^\dagger U = \hat{I}$ then for any u, v :

$$\langle Uu, Uv \rangle = \langle u, U^\dagger U v \rangle = \langle u, v \rangle. \quad (0.5.32)$$

The inner product is untouched. Setting $v = u$ gives $\|Uu\| = \|u\|$ — lengths preserved. Feeding both facts into (0.5.11) leaves $\cos \theta$ unchanged — angles preserved. And taking u, v from an orthonormal basis shows $\{Ue_i\}$ is again orthonormal: **unitary maps are exactly the maps that carry orthonormal bases to orthonormal bases.**

A real unitary matrix satisfies $U^\top U = I$, which is precisely an orthogonal matrix — a rotation or a reflection.

Unitary is the complex analogue of rotation. Hold that next to the last section of Chapter 0.3, where $e^{i\theta}$ turned out to be a rotation of the plane: the connection is not an analogy, and §7 will make it an identity.

Where this goes: in Chapter 4.4, the state of a quantum system evolves by $|\psi(t)\rangle = U(t) |\psi(0)\rangle$, and $U(t)$ must be unitary *because* $\|\psi\|^2 = 1$ is a total probability and probabilities must stay summing to one. (0.5.32) is that requirement, and it is the reason time evolution in quantum mechanics is a rotation rather than a flow that stretches.

5 · Eigenvalues and eigenvectors

A linear map generally does two things at once to a vector: turns it and rescales it. Some directions are special.

Definition. A non-zero v is an **eigenvector** of A with **eigenvalue** λ if

$$Av = \lambda v. \quad (0.5.33)$$

Geometrically: v spans a direction the map does not turn — it only stretches it, by the factor λ . (Negative λ flips it; complex λ means, as we will see, that no such real direction exists.) The requirement $v \neq 0$ is essential, since $A0 = \lambda 0$ holds for every λ and would make the definition vacuous.

To find them, rewrite (0.5.33) as $(A - \lambda \hat{I})v = 0$. A non-zero solution exists exactly when $A - \lambda \hat{I}$ fails to be invertible, which by Chapter 0.4 happens exactly when its determinant vanishes. So the eigenvalues are the roots of the **characteristic polynomial**:

$$p(\lambda) = \det(A - \lambda \hat{I}) = 0. \quad (0.5.34)$$

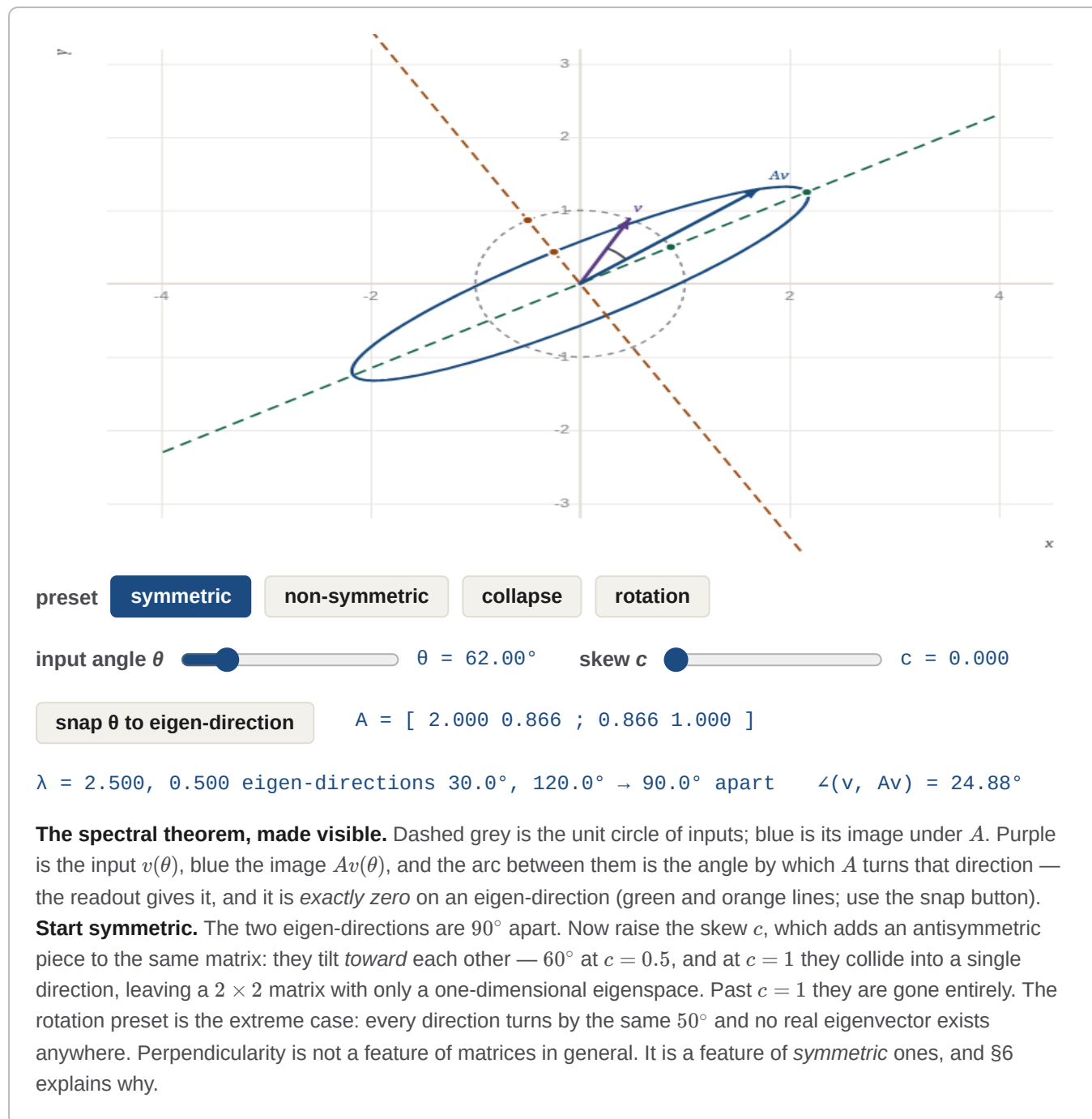
Expanding the determinant of an $n \times n$ matrix produces a polynomial of degree n in λ , with leading term $(-\lambda)^n$. Over $\mathbb{R}$ a polynomial need not have any roots ($\lambda^2 + 1$ has none), and correspondingly a real matrix can have no real eigenvectors at all — a rotation of the plane turns *every* direction, so none is preserved. Over $\mathbb{C}$ this cannot happen: as recorded in Chapter 0.4, every non-constant polynomial has a complex root, so

$$\text{every operator on a complex vector space has at least one eigenvalue.} \quad (0.5.35)$$

That single sentence is why quantum mechanics is built over $\mathbb{C}$ and not $\mathbb{R}$, and it is the hypothesis that makes §6 work. The set of eigenvectors for a given λ , together with 0, forms the **eigenspace** $E_\lambda = \ker(A - \lambda \hat{I})$, a subspace of dimension at least one. An eigenvalue whose eigenspace has dimension greater than one is called **degenerate**.

Seeing it

Before proving anything, look at the phenomenon. The figure below applies a 2×2 real matrix to every vector on the unit circle. The image is an ellipse — take that as observed for the moment; §6 will prove it, and the proof turns out to be the spectral theorem itself. A rotating input vector is drawn together with its image, and the eigen-directions — the angles at which input and image become parallel — are marked.



The figure's whole behaviour is one small computation. Write the family it uses as a symmetric matrix plus an antisymmetric one,

$$A(c) = \begin{pmatrix} a & b+c \\ b-c & d \end{pmatrix}, \quad \text{tr } A = a + d, \quad \det A = ad - b^2 + c^2, \quad (0.5.36)$$

and form the discriminant of the quadratic (0.5.34), whose roots are $\lambda = \frac{1}{2}(\text{tr } A \pm \sqrt{\mathcal{D}})$:

$$\mathcal{D} = (\text{tr } A)^2 - 4 \det A = (a - d)^2 + 4b^2 - 4c^2. \quad (0.5.37)$$

At $c = 0$ the discriminant is a sum of squares, so it is non-negative and the eigenvalues of a real symmetric 2×2 matrix are always real — a preview of §6(a). Turning on c eats into it, and the eigenvalues stay real only until $c^2 = \left(\frac{a-d}{2}\right)^2 + b^2$, which is exactly the squared half-gap between the eigenvalues of the symmetric part. In the figure $a = 2$, $b = \sqrt{3}/2$, $d = 1$, so the symmetric matrix has eigenvalues 2.5 and 0.5, half-gap 1, and the collapse happens at $c = 1$ — which is what the button does.

6 · The spectral theorem

This is the centre of the chapter and of Part 0. Everything before it was preparation, and a great deal after it is consequence.

(a) Hermitian $\Rightarrow$ every eigenvalue is real

Let $A^\dagger = A$ and $Av = \lambda v$ with $v \neq 0$. Since A is Hermitian, the defining property (0.5.28) reads $\langle Au, w \rangle = \langle u, Aw \rangle$. Apply it with $u = w = v$ and evaluate each side using (0.5.33):

$$\bar{\lambda} \langle v, v \rangle = \langle \lambda v, v \rangle = \langle Av, v \rangle = \langle v, Av \rangle = \langle v, \lambda v \rangle = \lambda \langle v, v \rangle. \quad (0.5.38)$$

The first step used conjugate-linearity in slot 1, the last linearity in slot 2. Since $v \neq 0$, axiom (iii) gives $\langle v, v \rangle > 0$, so we may divide:

$$\bar{\lambda} = \lambda \quad \implies \quad \lambda \in \mathbb{R}. \quad (0.5.39)$$

Three lines. The entire asymmetry between the two slots of the inner product has been converted into a statement that a number is real.

(b) Hermitian $\Rightarrow$ eigenvectors with distinct eigenvalues are orthogonal

Let $Av = \lambda v$ and $Aw = \mu w$ with $\lambda \neq \mu$, both real by (a). Compute $\langle v, Aw \rangle$ two ways:

$$\begin{aligned}
\langle v, Aw \rangle &= \langle v, \mu w \rangle = \mu \langle v, w \rangle, \\
\langle v, Aw \rangle &= \langle Av, w \rangle = \langle \lambda v, w \rangle = \bar{\lambda} \langle v, w \rangle = \lambda \langle v, w \rangle.
\end{aligned}
\tag{0.5.40}$$

The last step is where reality is spent: $\bar{\lambda} = \lambda$ by (a). Subtracting,

$$(\mu - \lambda) \langle v, w \rangle = 0 \quad \implies \quad \langle v, w \rangle = 0,$$
(0.5.41)

since $\mu - \lambda \neq 0$. Different eigenvalues force exactly perpendicular eigenvectors — not merely independent, *perpendicular*. Nothing was assumed about A but $A^\dagger = A$.

(c) The theorem

THE SPECTRAL THEOREM (FINITE-DIMENSIONAL, HERMITIAN)

Let A be a Hermitian operator on a finite-dimensional complex inner-product space V . Then V has an **orthonormal basis consisting of eigenvectors of A** , with real eigenvalues $\lambda_1, \dots, \lambda_n$, and

$$A = \sum_{i=1}^n \lambda_i |e_i\rangle \langle e_i|.$$

Before the proof, note what the last equation says. Compare it with completeness (0.5.21): the identity is $\sum_i |e_i\rangle \langle e_i|$, and A is the *same sum with the terms weighted by the eigenvalues*. An operator that would be complicated in a random basis is, in its own eigenbasis, nothing but a list of numbers attached to a list of perpendicular directions.

Granting the orthonormal eigenbasis, the formula is immediate: both sides are linear maps, and they agree on every basis vector, since

$$\left(\sum_i \lambda_i |e_i\rangle \langle e_i| \right) |e_j\rangle = \sum_i \lambda_i |e_i\rangle \delta_{ij} = \lambda_j |e_j\rangle = A |e_j\rangle,$$
(0.5.42)

and by Chapter 0.4 two linear maps agreeing on a basis are equal. So the work is entirely in producing the basis.

▼ Grind box — the induction proof of the spectral theorem

We induct on $n = \dim V$. The claim to be proved at each stage is: *every Hermitian operator on an n -dimensional complex inner-product space has an orthonormal basis of eigenvectors.*

Base case $n = 1$. Pick any unit vector e_1 ; it spans V , so Ae_1 is a multiple of it, $Ae_1 = \lambda_1 e_1$. Done, and λ_1 is real by part (a).

Step 1 — get one eigenvector. Let $\dim V = n \geq 2$ and assume the claim in dimension $n - 1$. Because V is complex, (0.5.35) supplies an eigenvalue λ_1 and an eigenvector, which we normalise to a unit vector e_1 . By (a), $\lambda_1 \in \mathbb{R}$.

Step 2 — pass to the orthogonal complement. Define

$$W = e_1^\perp = \{ w \in V : \langle e_1, w \rangle = 0 \}.$$

It is a subspace, because $\langle e_1, \cdot \rangle$ is linear. It inherits the inner product of V and every axiom still holds on a subset, so W is a complex inner-product space. We claim $\dim W = n - 1$. Split any $v \in V$ using the projection of §3 onto the line through e_1 :

$$v = \underbrace{\langle e_1, v \rangle e_1}_{\in \text{span}\{e_1\}} + \underbrace{(v - \langle e_1, v \rangle e_1)}_{\in W},$$

the second bracket lying in W by (0.5.25). So if $\{f_2, \dots, f_m\}$ is any basis of W , then $\{e_1, f_2, \dots, f_m\}$ spans V . It is also independent: applying $\langle e_1, \cdot \rangle$ to a vanishing combination $\alpha e_1 + \sum_j \beta_j f_j = 0$ kills every f_j term and leaves $\alpha = 0$, after which the independence of the f_j forces every $\beta_j = 0$. So it is a basis, and since dimension is well defined (Chapter 0.4), $1 + \dim W = n$.

Step 3 — A maps W into W . This is the step that uses Hermiticity, and it is the hinge of the whole proof. For $w \in W$,

$$\langle e_1, Aw \rangle = \langle Ae_1, w \rangle = \langle \lambda_1 e_1, w \rangle = \lambda_1 \langle e_1, w \rangle = 0,$$

using $A^\dagger = A$ in the first step and $\bar{\lambda}_1 = \lambda_1$ in the third. So $Aw \in W$: the restriction $A|_W$ is a genuine operator on W . *Without Hermiticity this fails* — a general operator does not preserve the orthogonal complement of one of its eigenvectors, and the induction dies right here. That is precisely why the figure in §5 loses its perpendicular eigen-directions the moment you break the symmetry.

Step 4 — the restriction is still Hermitian. For $u, w \in W$, the defining property is inherited verbatim, because it is a statement about inner products of vectors that all lie in W :

$$\langle A|_W u, w \rangle = \langle Au, w \rangle = \langle u, Aw \rangle = \langle u, A|_W w \rangle.$$

Step 5 — induct. By the induction hypothesis W has an orthonormal basis $e_2, \dots, e_n$ of eigenvectors of $A|_W$. But $A|_W$ acts as A does, so these are eigenvectors of A . Each lies in W , hence is orthogonal to e_1 ; and they are orthonormal among themselves. So $\{e_1, e_2, \dots, e_n\}$ is an orthonormal set of n eigenvectors in an n -dimensional space — a basis. ■

What was used. Exactly three things: that $\mathbb{C}$ guarantees an eigenvalue (0.5.35); that Hermiticity makes the orthogonal complement invariant; that the property survives restriction. Remove any one and the theorem is false.

Degeneracy, and the honest statement

Nothing above assumed the eigenvalues were distinct. If λ is degenerate, the induction simply hands you several basis vectors with the same eigenvalue, and *any* orthonormal basis of the eigenspace E_λ will do — the choice is not unique, and no theorem prefers one. What is unique is the eigenspace itself. Grouping the terms of the spectral decomposition by distinct eigenvalue λ_k and writing P_k for the projection onto E_{λ_k} :

$$A = \sum_k \lambda_k P_k, \quad P_k P_l = \delta_{kl} P_k, \quad \sum_k P_k = \hat{I}. \quad (0.5.43)$$

The middle identity holds because eigenvectors from different eigenspaces are orthogonal by (b); the right-hand one is completeness (0.5.21) regrouped. This is the basis-independent form, and it is the one that survives to infinite dimensions in Chapter 4.3.

Finally, the matrix version. Assemble the orthonormal eigenvectors as the columns of a matrix U . Orthonormality of the columns is exactly the statement $U^\dagger U = \hat{I}$, so U is unitary, and $Ae_i = \lambda_i e_i$ for every column reads

$$A = U D U^\dagger, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_i \in \mathbb{R}. \quad (0.5.44)$$

In the language of Chapter 0.4 this is a change of basis — and the change of basis is *unitary*, i.e. a rotation. **Every Hermitian operator is a diagonal one, seen from a rotated angle.** Two immediate dividends, using cyclicity of the trace from Chapter 0.4:

$$\text{tr } A = \text{tr}(D U^\dagger U) = \sum_i \lambda_i, \quad \det A = \prod_i \lambda_i. \quad (0.5.45)$$

The real case, and the ellipse we borrowed

A real symmetric matrix is Hermitian, so (a) gives it real eigenvalues. For a real λ the matrix $A - \lambda I$ is real and singular, so it has a real null vector; and Gram–Schmidt inside each eigenspace uses only real arithmetic. So **a real symmetric matrix has an orthonormal basis of real eigenvectors**, and (0.5.44) becomes $A = O D O^\top$ with O real orthogonal — the classical *principal axes theorem*.

That closes the loan taken out in §5. Let A be real, invertible, acting on the unit circle. A point x of the image is $x = Au$ with $\|u\| = 1$, i.e. $\|A^{-1}x\| = 1$, which squared reads

$$x^\top M x = 1, \quad M = (A^{-1})^\top A^{-1}. \quad (0.5.46)$$

Now $M^\top = M$, and $x^\top M x = \|A^{-1}x\|^2 > 0$ for $x \neq 0$. So M is real symmetric with strictly positive eigenvalues μ_i (each $\mu_i = u_i^\top M u_i > 0$), and substituting $x = O y$ turns (0.5.46) into $\sum_i \mu_i y_i^2 = 1$ — an ellipse with semi-axes $1/\sqrt{\mu_i}$ along the eigenvectors of M . The image of a circle under any invertible linear map is an ellipse, and the reason is the theorem you just proved.

One detail is worth extracting, because the figure in §5 shows it. The ellipse's axes are the eigenvectors of M , *not* of A . If $A = O D O^\top$ is symmetric then $M = O D^{-2} O^\top$ has the same eigenvectors and the two coincide — which is why, in

the symmetric preset, the eigen-lines lie exactly along the ellipse. Break the symmetry and they part company, visibly.

WHAT YOU HAVE JUST PROVED, IN PHYSICS LANGUAGE

Read the three results back, replacing "Hermitian operator" with "observable" and "eigenvalue" with "possible measurement outcome". Do not accept the replacement yet — just look at the sentences.

(a) Eigenvalues are real becomes: *measurement outcomes are real numbers*. An apparatus reads out a number on a dial; it cannot read $3 + 2i$ grams. In Chapter 4.2 this is presented as a requirement on the theory. It is not a requirement. It is (0.5.39).

(b) Eigenvectors are orthogonal becomes: *distinct outcomes are perfectly distinguishable*. Two states with different measured values have zero overlap, so the probability of confusing one for the other — which will turn out to be $|\langle u, v \rangle|^2$ — is exactly zero. Not small. Zero.

(c) The eigenvectors form a complete orthonormal basis becomes: *any state whatsoever can be written as a superposition of outcomes, $|\psi\rangle = \sum_i c_i |e_i\rangle$ with $c_i = \langle e_i | \psi \rangle$, and $\sum_i |c_i|^2 = \|\psi\|^2 = 1$ by Parseval (0.5.15) — the probabilities add to one automatically.*

These are three of the postulates of quantum mechanics as they are usually taught, and **they are not postulates. They are theorems about Hermitian matrices, proved above, in a chapter containing no physics.** Chapter 4.2 will do exactly one thing that this chapter did not: assert that a physical state is a unit vector and a physical observable is a Hermitian operator. That is one postulate, not four. Everything else is renaming.

△ WHY THIS ISN'T OBVIOUS

Most matrices are not diagonalisable at all. The spectral theorem's hypothesis is doing heavy work, and it is easy to miss how heavy. Consider

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \det(J - \lambda I) = (1 - \lambda)^2.$$

The only eigenvalue is $\lambda = 1$, repeated twice. Its eigenvectors solve $(J - I)v = 0$, i.e. $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v = 0$, which forces $v_2 = 0$: the eigenspace is the single line spanned by $(1, 0)$. A two-dimensional space, and only a one-dimensional supply of eigenvectors. There is no basis of eigenvectors, no diagonal form, and no spectral decomposition. Such a matrix is called **defective**, and it is not exotic — the skew slider in the figure above manufactures one at $c = 1$. Of course $J^\dagger \neq J$; the theorem is not violated, it is simply inapplicable. When a physics text says "expand in eigenstates", it is quietly leaning on Hermiticity, every time.

In infinite dimensions the theorem needs genuine repair. Everything proved in this chapter is a *finite-dimensional* theorem, and two of the steps fail outright in a function space. First, "Hermitian" and "self-adjoint" come apart: an unbounded operator carries a *domain* — the subset of functions it is allowed to act on — and $A^\dagger$ carries its own, generally larger. The symmetric condition $\langle Au, v \rangle = \langle u, Av \rangle$ on a domain is strictly weaker than genuine self-adjointness, and the difference is not pedantry: the momentum operator $-i\hbar d/dx$ on the half-line $[0, \infty)$ is symmetric and admits *no* self-adjoint extension whatsoever. Second, eigenvectors can fail to exist inside the space at all: on the whole line, the momentum eigenfunctions e^{ikx} solve the eigenvalue equation but are not square-integrable, so they are not vectors of the Hilbert space. The spectrum becomes *continuous*, the sum $\sum_k \lambda_k P_k$ becomes an integral $\int \lambda dP(\lambda)$ over projection-valued measures, and completeness must be re-proved rather than assumed. **Chapter 4.3 pays this bill in full.** Until then, everything here is exact and everything here is finite-dimensional.

7 · Functions of operators

You know what e^x means for a number. What could e^A mean for an operator? The spectral theorem answers it at once. If A acts on each e_i by multiplying by λ_i , then whatever $f(A)$ is, it ought to act on e_i by multiplying by $f(\lambda_i)$. So define

$$f(A) \equiv \sum_i f(\lambda_i) |e_i\rangle \langle e_i| \quad (0.5.47)$$

for A Hermitian with spectral decomposition $A = \sum_i \lambda_i |e_i\rangle \langle e_i|$, and any function f defined on the eigenvalues. In the projection form (0.5.43) this is $f(A) = \sum_k f(\lambda_k) P_k$, which makes the definition manifestly independent of which orthonormal basis you chose inside a degenerate eigenspace: the P_k are unique. So (0.5.47) is well posed.

It had better agree with the answer you would get by substituting A into a power series, and it does. The key is that orthonormality makes the cross terms vanish:

$$A^2 = \sum_{i,j} \lambda_i \lambda_j |e_i\rangle \underbrace{\langle e_i | e_j \rangle}_{\delta_{ij}} \langle e_j| = \sum_i \lambda_i^2 |e_i\rangle \langle e_i|, \quad (0.5.48)$$

and by the same collapse, inductively, $A^m = \sum_i \lambda_i^m |e_i\rangle \langle e_i|$ for every $m \geq 1$. It holds for $m = 0$ as well, since $A^0 = \hat{I} = \sum_i |e_i\rangle \langle e_i|$ by completeness. Therefore, for any $f(z) = \sum_m c_m z^m$ whose series converges at every eigenvalue,

$$\sum_m c_m A^m = \sum_i \left(\sum_m c_m \lambda_i^m \right) |e_i\rangle \langle e_i| = \sum_i f(\lambda_i) |e_i\rangle \langle e_i|. \quad (0.5.49)$$

Swapping the two sums is legitimate because the sum over i is finite. So (0.5.47) and "plug the matrix into the series" agree wherever both make sense — and (0.5.47) keeps working for functions with no power series, such as $\sqrt{\cdot}$ on a positive operator.

The exponential of i times a Hermitian operator

Claim: if A is Hermitian then $U = e^{iA}$ is unitary.

Take the adjoint of (0.5.47) with $f(\lambda) = e^{i\lambda}$, using $(\alpha M)^\dagger = \bar{\alpha} M^\dagger$ and $(|e\rangle \langle e|)^\dagger = |e\rangle \langle e|$ from (0.5.31):

$$(e^{iA})^\dagger = \sum_i \overline{e^{i\lambda_i}} |e_i\rangle \langle e_i| = \sum_i e^{-i\lambda_i} |e_i\rangle \langle e_i| = e^{-iA}. \quad (0.5.50)$$

The middle step is where reality is spent: $\overline{e^{i\lambda}} = e^{-i\bar{\lambda}}$, and $\bar{\lambda} = \lambda$ only because A is Hermitian. Now multiply, collapsing with orthonormality exactly as in (0.5.48):

$$(e^{iA})^\dagger e^{iA} = \sum_i e^{-i\lambda_i} e^{i\lambda_i} |e_i\rangle \langle e_i| = \sum_i |e_i\rangle \langle e_i| = \hat{I}. \quad (0.5.51)$$

So e^{iA} is unitary. ■ And the converse holds too: every unitary operator is e^{iA} for some Hermitian A (grind box).

Hermitian operators and unitary operators are the same information, related by an exponential — exactly as real numbers and points on the unit circle are related by $\theta \mapsto e^{i\theta}$ in Chapter 0.3. This is that statement with the number replaced by an operator.

▼ Grind box — the converse: every unitary is e^{iA}

We need a spectral theorem for unitary operators. The induction of §6 goes through with two substitutions.

Eigenvalues have modulus one. If $U|v\rangle = \mu|v\rangle$ with $v \neq 0$ then by (0.5.32), $\langle v, v \rangle = \langle Uv, Uv \rangle = |\mu|^2 \langle v, v \rangle$, so $|\mu| = 1$ and we may write $\mu = e^{i\theta}$ with θ real.

The orthogonal complement is invariant. In finite dimensions $U^\dagger = U^{-1}$, and $Uv = \mu v$ gives $U^\dagger v = \mu^{-1} v = \bar{\mu} v$ since $|\mu| = 1$. So for $w \perp v$,

$$\langle v, Uw \rangle = \langle U^\dagger v, w \rangle = \langle \bar{\mu} v, w \rangle = \mu \langle v, w \rangle = 0,$$

so $Uw \perp v$. The restriction $U|_W$ preserves inner products on W and maps W into W ; being injective on a finite-dimensional space it is onto, hence unitary on W .

Induct exactly as before, using (0.5.35) to start. The result is an orthonormal basis $\{f_i\}$ with $Uf_i = e^{i\theta_i} f_i$, so

$$U = \sum_i e^{i\theta_i} |f_i\rangle \langle f_i| = e^{iA}, \quad A \equiv \sum_i \theta_i |f_i\rangle \langle f_i|.$$

That A is Hermitian because the θ_i are real and each $|f_i\rangle \langle f_i|$ is self-adjoint. ■ (The θ_i are only defined modulo 2π , so A is not unique — the exponential map is many-to-one, exactly as $e^{i\theta}$ is on $\mathbb{R}$.)

WHERE THIS IS SPENT

In Chapter 4.4 the Hamiltonian $\hat{H}$ is Hermitian because energy is an observable, and time evolution is

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(0)\rangle.$$

That operator is unitary *by the theorem just proved*, and unitary means probability is conserved by (0.5.32). So the sentence "energy is observable, therefore probability is conserved" is not a physical argument at all — it is (0.5.51). Differentiate the exponential and you get $i\hbar d|\psi\rangle/dt = \hat{H}|\psi\rangle$: the Schrödinger equation is the statement that a Hermitian operator generates a unitary flow.

This pairing — *Hermitian generator, unitary group element* — is the finite-dimensional shadow of the Lie algebra / Lie group relationship of Chapter 6.1, where $\mathfrak{e}$ maps the algebra (a vector space of generators) to the group (a curved space of transformations). Every symmetry in the Standard Model is an instance. You have just met the whole idea in two dimensions, with no differential geometry anywhere.

8 · Commuting operators and simultaneous diagonalisation

Define the **commutator** $[A, B] \equiv AB - BA$. It measures the failure of two operators to be interchangeable, and it is zero exactly when order does not matter. The theorem below is the reason that quantity is worth a name.

Theorem. Two Hermitian operators A and B on a finite-dimensional complex inner-product space admit a common orthonormal basis of eigenvectors if and only if $[A, B] = 0$.

Forward direction (easy)

Suppose $\{e_i\}$ is an orthonormal basis with $Ae_i = a_i e_i$ and $Be_i = b_i e_i$. Then

$$ABe_i = A(b_i e_i) = a_i b_i e_i, \quad BAe_i = B(a_i e_i) = b_i a_i e_i, \quad (0.5.52)$$

and $a_i b_i = b_i a_i$ because they are numbers. So AB and BA agree on a basis, hence are equal (Chapter 0.4), so $[A, B] = 0$. ■

Reverse direction (the real content)

Here the difficulty is degeneracy. If A had n distinct eigenvalues the argument would be three lines. It generally does not, and the honest proof must build the common basis inside each eigenspace.

▼ Grind box — the reverse direction, degeneracy handled

Assume $A^\dagger = A$, $B^\dagger = B$, and $AB = BA$.

Step 1 — decompose V by A . By the spectral theorem, V is the orthogonal direct sum of the eigenspaces of A for its distinct eigenvalues $\lambda_1, \dots, \lambda_r$:

$$V = E_{\lambda_1} \oplus \dots \oplus E_{\lambda_r}, \quad E_{\lambda_k} \perp E_{\lambda_l} \ (k \neq l),$$

the orthogonality being part (b) of §6 and the spanning being part (c).

Step 2 — B preserves each eigenspace. This is the only place the hypothesis $[A, B] = 0$ is used. Let $v \in E_{\lambda_k}$, so $Av = \lambda_k v$. Then

$$A(Bv) = (AB)v = (BA)v = B(Av) = B(\lambda_k v) = \lambda_k (Bv),$$

so Bv is again an eigenvector of A with eigenvalue λ_k (or zero, which lies in E_{λ_k} too). Hence $B(E_{\lambda_k}) \subseteq E_{\lambda_k}$.

Step 3 — the restriction is Hermitian. Exactly as in Step 4 of the spectral-theorem proof: for $u, w \in E_{\lambda_k}$, $\langle Bu, w \rangle = \langle u, Bw \rangle$ holds because it holds in all of V , and both vectors lie in the subspace. So $B|_{E_{\lambda_k}}$ is a Hermitian operator on the finite-dimensional inner-product space E_{λ_k} .

Step 4 — apply the spectral theorem inside the eigenspace. Therefore E_{λ_k} has an orthonormal basis of eigenvectors of B . And every vector in E_{λ_k} is automatically an eigenvector of A , with eigenvalue λ_k — that is what the eigenspace is. So this basis consists of simultaneous eigenvectors of both operators. *This is exactly where degeneracy is handled:* within a degenerate eigenspace of A , the operator B chooses the basis that A could not.

Step 5 — assemble. Take the union of these bases over $k = 1, \dots, r$. Vectors from the same eigenspace are orthonormal by construction; vectors from different eigenspaces are orthogonal by Step

1. The total count is $\sum_k \dim E_{\lambda_k} = \dim V$, so the union is an orthonormal basis of V consisting of simultaneous eigenvectors of A and B . ■

Remark. The same argument run with three or more mutually commuting Hermitian operators produces a basis of simultaneous eigenvectors for all of them — apply Step 2 to each in turn, refining the decomposition until no eigenspace has dimension greater than one.

THIS THEOREM IS THE STRUCTURE OF QUANTUM NUMBERS

An orthonormal basis of simultaneous eigenvectors is a way of *labelling* every state by a list of numbers — one eigenvalue from each operator. Chapter 4.7 calls such a list the **quantum numbers** of the state, and the operators a *complete set of commuting observables*: mutually commuting, and enough of them that every eigenspace has been cut down to a single line, so the labels determine the state uniquely. When you write the hydrogen states as $|n, \ell, m\rangle$ you are naming the eigenvalues of three commuting operators, and the reason the labels are consistent — the reason a state can have a definite ℓ *and* a definite m at the same time — is Step 4 above, and nothing else.

Run it backwards and you get the other half. If $[A, B] \neq 0$ there is *no* common eigenbasis, so there is no state in which both observables are sharp. That is the qualitative content of Chapter 4.6, and the quantitative version is the Cauchy–Schwarz bound $\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$ from §1, whose right-hand side is non-zero exactly when the operators fail to commute. The two great mysteries of quantum mechanics — *why can some quantities be known together and others not?* — are one theorem about matrices, proved above, with an "if and only if" in the middle.

FAMILIAR GROUND

You have run principal component analysis on expression data, on imaging features, on multiplexed assay panels. Here is what was actually happening, and why it could not have failed.

Centre the data matrix X (n samples $\times$ p features) and form the covariance matrix $\Sigma = \frac{1}{n-1} X^\top X$, whose entries are $\Sigma_{ij} = \text{cov}(\text{feature } i, \text{feature } j)$. Covariance is symmetric in its two arguments, so $\Sigma^\top = \Sigma$ — **a real covariance matrix is Hermitian by construction**, not by assumption or by luck. Section 6 therefore *guarantees*, as a theorem:

- **The eigenvalues are real.** They are variances, so this had better be true. It is not a numerical accident, and it does not depend on the data.
- **The principal components are mutually orthogonal.** Uncorrelated components are not something PCA arranges by cleverness. They come out perpendicular because §6(b) says eigenvectors of a symmetric matrix with distinct eigenvalues have no choice.
- **They span the whole feature space.** Completeness (0.5.21) says no direction of variation is missed, so reconstructing from all p components returns your data exactly.

More is true, and it is worth being precise about the extra. Σ is not merely Hermitian, it is **positive semi-definite**: for any direction u ,

$$u^\top \Sigma u = \frac{1}{n-1} \|Xu\|^2 \geq 0,$$

because it is a sum of squares — and it is exactly the sample variance of the data projected onto u . Applying this to a unit eigenvector gives $\lambda = u^\top \Sigma u \geq 0$: the eigenvalues are not just real, they are non-negative, and *each eigenvalue is the variance along its own component*. Positive semi-definiteness is strictly stronger than Hermiticity — a Hermitian matrix may have negative eigenvalues, a covariance matrix may not.

Two further standard facts now cost one line each. Writing a unit vector as $u = \sum_i c_i u_i$ in the eigenbasis with $\sum_i c_i^2 = 1$, the spectral decomposition gives $u^\top \Sigma u = \sum_i \lambda_i c_i^2$, a weighted average of the eigenvalues — maximised by putting all the weight on the largest. *The first principal component is the top eigenvector because a weighted average is largest when it is not an average.* And by (0.5.45) the total variance $\text{tr } \Sigma = \sum_i \lambda_i$, which is why "proportion of variance explained" is $\lambda_i / \sum_j \lambda_j$ and why those fractions sum to one.

Now the point worth carrying away. **PCA works for exactly the same reason quantum measurement outcomes are real numbers.** One theorem, §6, applied twice: once to a covariance matrix, where the real eigenvalues are variances and the orthogonal eigenvectors are uncorrelated components; once to a Hamiltonian, where the real eigenvalues are energies and the orthogonal eigenvectors are distinguishable states. It is not an analogy and not a coincidence of formalism. It is one theorem with two costumes, and you have been using the harder-looking costume for years without noticing.

9 • Worked examples

WORKED EXAMPLE 1 — THE TWO-STATE SYSTEM, DIAGONALISED COMPLETELY

Take the Hermitian matrix

$$H = \begin{pmatrix} E_0 & \Delta \\ \Delta & E_0 \end{pmatrix}, \quad E_0, \Delta \in \mathbb{R}, \Delta > 0.$$

It is real and symmetric, hence Hermitian, so §6 applies before we compute anything: the eigenvalues will be real and the eigenvectors perpendicular. Let us watch it happen.

Eigenvalues. From (0.5.34),

$$\begin{aligned} \det \begin{pmatrix} E_0 - \lambda & \Delta \\ \Delta & E_0 - \lambda \end{pmatrix} &= (E_0 - \lambda)^2 - \Delta^2 = 0, \\ \implies E_0 - \lambda &= \pm \Delta \implies \lambda_{\pm} = E_0 \pm \Delta. \end{aligned}$$

Real, as promised, and note what has happened: a matrix with equal diagonal entries — two "levels" of the same value E_0 — has eigenvalues that are *split apart* by the off-diagonal coupling, into $E_0 + \Delta$ and $E_0 - \Delta$, separated by 2Δ .

Eigenvectors. For $\lambda_+ = E_0 + \Delta$, solve $(H - \lambda_+ I)v = 0$:

$$\begin{pmatrix} -\Delta & \Delta \\ \Delta & -\Delta \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = v_2 \implies |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_- = E_0 - \Delta$ the matrix becomes $\begin{pmatrix} \Delta & \Delta \\ \Delta & \Delta \end{pmatrix}$, forcing $v_1 = -v_2$, so $|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Both have been normalised by dividing by $\sqrt{1^2 + 1^2}$.

Orthogonality, checked. $\langle + | - \rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0$. ✓ Perpendicular, exactly as §6(b) requires — and we never used that fact to find them, so this is a genuine test.

Spectral decomposition. Build the projections $P_{\pm} = |\pm\rangle \langle \pm|$ as column times row:

$$P_+ = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \quad 1) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_- = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}.$$

Sanity checks: $P_+ + P_- = I$ ✓ (completeness), $P_+ P_- = 0$ ✓ (orthogonality), $P_{\pm}^2 = P_{\pm}$ ✓ (idempotence).

Reconstruction. Now rebuild H from its spectrum alone:

$$\begin{aligned} \lambda_+ P_+ + \lambda_- P_- &= \frac{E_0 + \Delta}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{E_0 - \Delta}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} E_0 & \Delta \\ \Delta & E_0 \end{pmatrix} = H. \quad \checkmark \end{aligned}$$

The matrix has been completely dissolved into two numbers and two perpendicular directions, and reassembled from them.

Why this example and not another. The eigenvalues $E_0 \pm \Delta$ are a *level splitting*: two states that would be degenerate at energy E_0 are pushed apart by the coupling Δ , and the new eigenstates are the symmetric and antisymmetric combinations rather than either original state. This 2×2 matrix is the most reused object in all of quantum mechanics. It is the ammonia molecule, whose two mirror-image configurations split by 2Δ and give the ammonia maser; it is neutrino oscillation, where the states of definite mass are not the states of definite flavour and the mismatch makes them convert into one another; it is the qubit, where $|+\rangle$ and $|-\rangle$ are the two states you engineer and control. Chapter 4.5 will do all three, and the linear algebra will already be finished.

WORKED EXAMPLE 2 — $E^{i\theta\sigma_x}$, OR EULER'S FORMULA WITH A MATRIX IN THE EXPONENT

Let

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

which is real symmetric, hence Hermitian. Compute $e^{i\theta\sigma_x}$ for real θ using (0.5.47) rather than by summing a series.

Step 1 — the spectrum. $\det(\sigma_x - \lambda I) = \lambda^2 - 1 = 0$, so $\lambda = \pm 1$. The eigenvectors are the same $|\pm\rangle$ as in Worked Example 1 (set $E_0 = 0, \Delta = 1$ there and compare), giving the spectral decomposition

$$\sigma_x = (+1)P_+ + (-1)P_- = P_+ - P_-.$$

Step 2 — apply the function. The operator in the exponent is $\theta\sigma_x$, whose eigenvalues are $\pm\theta$ on the same eigenvectors. So with $f(\lambda) = e^{i\lambda}$,

$$e^{i\theta\sigma_x} = e^{i\theta}P_+ + e^{-i\theta}P_-.$$

Step 3 — regroup using Euler. Write $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$ from Chapter 0.3 and collect the P 's:

$$\begin{aligned} e^{i\theta\sigma_x} &= \cos \theta (P_+ + P_-) + i \sin \theta (P_+ - P_-) \\ &= \cos \theta I + i \sin \theta \sigma_x, \end{aligned}$$

using completeness for the first bracket and Step 1 for the second. Explicitly,

$$e^{i\theta\sigma_x} = \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}.$$

What this is. Compare with $e^{i\theta} = \cos \theta + i \sin \theta$. The structure is identical — *this is Euler's formula with a matrix in the exponent* — and the reason it works is that $\sigma_x^2 = I$, so σ_x plays the role that i plays for numbers except that it squares to $+1$ rather than -1 ; the eigenvalues ± 1 are what carry the alternating signs. Any Hermitian M with $M^2 = I$ gives $e^{i\theta M} = \cos \theta I + i \sin \theta M$ by the identical three steps.

Checks. Unitarity: $(e^{i\theta\sigma_x})^\dagger = \cos \theta I - i \sin \theta \sigma_x$, and multiplying gives $\cos^2 \theta I + \sin^2 \theta \sigma_x^2 = I \checkmark$, as §7 promised. Determinant: $\cos^2 \theta - (i \sin \theta)^2 = \cos^2 \theta + \sin^2 \theta = 1 \checkmark$.

Where it goes. Unitary 2×2 matrices of determinant 1 form the group $SU(2)$, and the operator above is a member of it. In Chapter 4.7, $e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$ is precisely the operator that rotates a spin- $\frac{1}{2}$ state about the axis $\hat{n}$ by angle θ — the same computation, with a general unit vector in place of the x -direction. The factor of $\frac{1}{2}$ in the exponent is why a spin must be turned through 720° to return to itself, a fact you can already see coming: at $\theta = 2\pi$ the formula above gives $\cos \pi I = -I$, not I . Chapter 6.1 explains what kind of object does that.

10 · Your turn

PROBLEM 1 — GRAM-SCHMIDT, TWICE

(a) Apply (0.5.16) to the basis $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$, $v_3 = (0, 1, 1)$ of $\mathbb{R}^3$ with the ordinary dot product. Verify the result is orthonormal.

(b) Now do the same in a space with no arrows in it. Take the polynomials $1, x, x^2$ on $[-1, 1]$ with $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$, and orthonormalise them.

– Solution

(a) $\|v_1\| = \sqrt{2}$, so $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$.

Next, $\langle e_1, v_2 \rangle = \frac{1}{\sqrt{2}}(1 + 0 + 0) = \frac{1}{\sqrt{2}}$, so

$$w_2 = (1, 0, 1) - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, 1, 0) = \left(\frac{1}{2}, -\frac{1}{2}, 1\right),$$

$$\|w_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}} \implies e_2 = \frac{1}{\sqrt{6}}(1, -1, 2).$$

Finally $\langle e_1, v_3 \rangle = \frac{1}{\sqrt{2}}$ and $\langle e_2, v_3 \rangle = \frac{1}{\sqrt{6}}(0 - 1 + 2) = \frac{1}{\sqrt{6}}$, so

$$\begin{aligned} w_3 &= (0, 1, 1) - \frac{1}{2}(1, 1, 0) - \frac{1}{6}(1, -1, 2) \\ &= \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right), \quad \|w_3\| = \frac{2}{\sqrt{3}}, \end{aligned}$$

$$e_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

Check. $\langle e_1, e_2 \rangle = \frac{1}{\sqrt{12}}(1 - 1 + 0) = 0$; $\langle e_1, e_3 \rangle = \frac{1}{\sqrt{6}}(-1 + 1 + 0) = 0$; $\langle e_2, e_3 \rangle = \frac{1}{\sqrt{18}}(-1 - 1 + 2) = 0$; and each has norm 1 by construction. ✓

(b) $\langle 1, 1 \rangle = \int_{-1}^1 dx = 2$, so $e_0 = 1/\sqrt{2}$.

$\langle e_0, x \rangle = \frac{1}{\sqrt{2}} \int_{-1}^1 x dx = 0$ — the integrand is odd, so x is *already* orthogonal to the constants and nothing needs subtracting. With $\langle x, x \rangle = \int_{-1}^1 x^2 dx = \frac{2}{3}$,

$$e_1 = \sqrt{\frac{3}{2}} x.$$

For x^2 : $\langle e_1, x^2 \rangle = 0$ by oddness again, while $\langle e_0, x^2 \rangle = \frac{1}{\sqrt{2}} \cdot \frac{2}{3}$. Subtracting,

$$w_2 = x^2 - \frac{1}{\sqrt{2}} \cdot \frac{2}{3} \cdot \frac{1}{\sqrt{2}} = x^2 - \frac{1}{3},$$

$$\|w_2\|^2 = \int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45},$$

$$e_2 = \sqrt{\frac{45}{8}} \left(x^2 - \frac{1}{3}\right) = \sqrt{\frac{5}{2}} \cdot \frac{3x^2 - 1}{2}.$$

Those are the first three **Legendre polynomials**, normalised: $P_0 = 1$, $P_1 = x$, $P_2 = (3x^2 - 1)/2$, each divided by $\sqrt{2/(2n+1)}$. They were not invented; they are what you get by running Gram–Schmidt on $1, x, x^2, \dots$ and there is nothing else they could have been. The identical procedure with a weight e^{-x^2} on the whole line produces the Hermite polynomials, which are the quantum harmonic oscillator states of Chapter 4.5. *The machinery does not care that the vectors are functions.* That is the whole content of the abstraction, and it is what Chapter 4.3 will exploit.

PROBLEM 2 — THE SPECTRUM OF A UNITARY OPERATOR

Prove that every eigenvalue of a unitary operator has $|\lambda| = 1$. Then interpret it: what does this say geometrically, and what would go wrong physically if it were false?

– Solution

Let $Uv = \lambda v$ with $v \neq 0$. Apply (0.5.32) with both arguments equal to v :

$$\langle v, v \rangle = \langle Uv, Uv \rangle = \langle \lambda v, \lambda v \rangle = \bar{\lambda}\lambda \langle v, v \rangle = |\lambda|^2 \langle v, v \rangle.$$

Since $v \neq 0$, axiom (iii) gives $\langle v, v \rangle > 0$ and we may divide: $|\lambda|^2 = 1$, so $|\lambda| = 1$ and $\lambda = e^{i\varphi}$ for some real φ .

Geometry. A unitary map preserves all lengths, so along any direction it preserves it must stretch by a factor of modulus one — i.e. not stretch at all, only rotate the phase. A unitary operator has no expanding or contracting directions anywhere; in its eigenbasis it is $\text{diag}(e^{i\varphi_1}, \dots, e^{i\varphi_n})$, a separate rotation on each of n perpendicular axes. Contrast a Hermitian operator, which in its eigenbasis is $\text{diag}(\lambda_1, \dots, \lambda_n)$ with the λ_i real: it *only* stretches and never rotates. The two classes are exactly complementary, which is the content of $U = e^{iA}$.

Physically. If some $|\lambda| > 1$, a state along that eigenvector would grow in norm under repeated application of U . Since $\|\psi\|^2$ is total probability, that state would end up with probability greater than one — an outcome would occur more often than always. If $|\lambda| < 1$ the state would fade away and probability would leak out of the universe. Chapter 4.4's insistence on unitarity is exactly the demand that neither happens, and by the result above there is no third option: unitary forces $|\lambda| = 1$ automatically.

PROBLEM 3 — SYMMETRY IS WHAT MAKES THE RIGHT ANGLE

- (a) Diagonalise $S = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$: find the eigenvalues, normalise the eigenvectors, and verify they are orthogonal.
- (b) Now break the symmetry with $N = \begin{pmatrix} 3 & \frac{3}{2} \\ \frac{1}{2} & 3 \end{pmatrix}$, which has the same symmetric part. Find its eigenvalues and eigen-directions and compute the angle between them. Comment on what happened.

– Solution

(a) $\det(S - \lambda I) = (3 - \lambda)^2 - 1 = 0$ gives $3 - \lambda = \pm 1$, so $\lambda = 4$ and $\lambda = 2$. For $\lambda = 4$: $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0$ forces $v_1 = v_2$, giving $u_1 = \frac{1}{\sqrt{2}}(1, 1)$, at 45° . For $\lambda = 2$: $v_1 = -v_2$, giving $u_2 = \frac{1}{\sqrt{2}}(1, -1)$, at -45° .

Orthogonality: $u_1 \cdot u_2 = \frac{1}{2}(1 - 1) = 0 \checkmark$. The two eigen-directions are exactly 90° apart, as §6(b) guarantees for any real symmetric matrix with distinct eigenvalues.

(b) $\text{tr } N = 6$ and $\det N = 9 - \frac{3}{4} = \frac{33}{4}$, so

$$\lambda = \frac{6 \pm \sqrt{36 - 33}}{2} = 3 \pm \frac{\sqrt{3}}{2} \approx 3.866, 2.134.$$

Real — but that is luck, not a theorem; the discriminant came out positive here and will not in general. For $\lambda_+ = 3 + \frac{\sqrt{3}}{2}$, the first row of $N - \lambda_+ I$ gives

$$-\frac{\sqrt{3}}{2}v_1 + \frac{3}{2}v_2 = 0 \implies v_2 = \frac{1}{\sqrt{3}}v_1,$$

a direction at $\arctan(1/\sqrt{3}) = 30^\circ$. For λ_- the sign flips and the direction is at -30° . Their dot product, with $v_\pm = (1, \pm 1/\sqrt{3})$, is $1 - \frac{1}{3} = \frac{2}{3} \neq 0$, and

$$\cos \alpha = \frac{2/3}{(2/\sqrt{3})^2} = \frac{2/3}{4/3} = \frac{1}{2} \implies \alpha = 60^\circ.$$

Comment. The right angle is gone: 90° became 60° . Nothing was done to the symmetric part of the matrix — $N = S + \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, an antisymmetric addition — and yet the eigen-directions tilted toward each other by 15° each. This is the figure of §5 with different numbers, and by (0.5.37) the eigenvalues would have gone complex once the antisymmetric coefficient exceeded half the eigenvalue gap of S , i.e. $c = 1$. Orthogonality of eigenvectors is not a property of matrices. It is a property of *Hermitian* matrices, and it is lost the moment the hypothesis is.

PROBLEM 4 — NON-DEGENERATE COMMUTING OPERATORS SHARE EVERYTHING

Let A and B be Hermitian with $[A, B] = 0$, and suppose A is *non-degenerate*: every eigenvalue of A has a one-dimensional eigenspace. Show that every eigenvector of A is automatically an eigenvector of B . Then explain why the general proof of §8 needed to be longer than this one.

– Solution

Let $Av = \lambda v$ with $v \neq 0$. Using $AB = BA$,

$$A(Bv) = (AB)v = (BA)v = B(Av) = B(\lambda v) = \lambda(Bv).$$

So Bv lies in the eigenspace E_λ . By hypothesis E_λ is one-dimensional and contains v , so $E_\lambda = \text{span}\{v\}$ and therefore $Bv = \mu v$ for some scalar μ . Hence v is an eigenvector of B . (If $Bv = 0$ then $\mu = 0$, which is still an eigenvalue — the zero vector is excluded as an *eigenvector*, not as an *eigenvalue*.) Since the eigenvectors of A already form an orthonormal basis, that basis diagonalises B too. ■

Why the general case is harder. The step " E_λ is one-dimensional, therefore Bv is a multiple of v " is exactly what fails under degeneracy. If $\dim E_\lambda = 3$, all we learn is that B maps that three-dimensional space into itself — Bv could be any vector in it, and a generic $v \in E_\lambda$ will *not* be an eigenvector of B . The repair, and the whole content of the grind box in §8, is that B restricted to E_λ is still Hermitian, so the spectral theorem applies *inside* E_λ and selects the right basis there. Degeneracy does not break the theorem; it just means A alone no longer specifies the basis and needs B 's help to finish the job.

The physics of that last sentence, for later: a degenerate energy level is one where the Hamiltonian alone does not tell you which state you are in, and you must measure a second commuting observable to say. That is exactly why hydrogen states need three labels rather than one, and why "lifting a degeneracy" — with a magnetic field, say — is such a common experimental move. Chapter 4.7.

THE BRICK YOU JUST LAID

You have the *inner product* from three axioms, with the conjugate forced by the demand that no non-zero vector have zero length, and every geometric notion — length, distance, orthogonality, angle — reconstructed from it. You have *Cauchy–Schwarz*, proved by optimising a manifestly non-negative quantity, which is what makes "angle" definable at all. You have *orthonormal bases*, in which coordinates are inner products; *Gram–Schmidt*, which shows they always exist; and *completeness*, $\sum_i |e_i\rangle \langle e_i| = \hat{I}$, in Dirac notation introduced where it is nothing but bookkeeping about rows and columns. You have *orthogonal projection* as the nearest-point map, and with it the fact that least-squares regression is a right angle and ANOVA is Pythagoras. You have the *adjoint* defined without reference to any basis, and shown to be the conjugate transpose in an orthonormal one. And you have the *spectral theorem*: a Hermitian operator has real eigenvalues, perpendicular eigenvectors, and a complete orthonormal eigenbasis, so that it dissolves into $\sum_i \lambda_i |e_i\rangle \langle e_i|$ — with functions of operators, the Hermitian–unitary correspondence $U = e^{iA}$, and simultaneous diagonalisation of commuting operators as corollaries.

Where this gets spent. *Cauchy–Schwarz* → Chapter 4.6, where it becomes the uncertainty principle with nothing added but interpretation. *Orthonormal bases and completeness* → Chapter 4.2 (superposition and probability amplitudes) and Chapter 0.9, where Fourier analysis turns out to be an orthonormal expansion in a function space and Parseval becomes conservation of energy across modes. *Adjoint and Hermiticity* → Chapter 4.2, where "an observable is a Hermitian operator" is the one postulate that has to be made. *The spectral theorem* → Chapter 4.2 (the measurement postulates, which are these theorems renamed), Chapter 4.5 (solving a system means diagonalising its Hamiltonian), Chapter 0.8 (normal modes of coupled oscillators are eigenvectors of a symmetric matrix), and every PCA you will ever run. e^{iA} → Chapter 4.4, where a Hermitian Hamiltonian generates unitary time evolution and probability is conserved as a matter of algebra, and Chapter 6.1, where the same pairing becomes the Lie algebra of a Lie group. *Commuting observables* → Chapters 4.6 and 4.7, where they become the complete set of quantum numbers that labels every state of every atom.

One thing remains outstanding, and it is worth naming so that you notice when it is paid. Every proof in this chapter used *finite dimension* — in the induction, in rank–nullity, in the interchange of sums, in the claim that an injective map is surjective. Quantum mechanics happens in infinite dimensions. Chapter 4.3 is where the bill comes due.